Performance of Matrix transpose with Cache-blocking

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Blocking is also called tiling.
Cache blocking is NOT same as a cache block.

With cache blocking
A in Row-major
B in Row-major

Code after doing cache-blocking

```
for (u =0; u < N; u+=Q) {
 for (v =0; v < N; v+=Q) {
  for (w=0; w<Q; w++) {
   for (x=0; x<Q; x++) {
    B[v+x][u+w] = A[u+w][v+x];
   }
  }
 }
}
```

With cache-blocking, B in row-major

B00	B01	B02	B03
B10	B11	B12	B13
B20	B21	B22	B23
B30	B21	B22	B23

A00	A01	A02	A03
A10	A11	A12	A13
A20	A21	A22	A23
A30	A21	A22	A23

```
for (u =0; u < N; u+=Q) {
 for (v =0; v < N; v+=Q) {
  for (w=0; w<Q; w++) {
   for (x=0; x<Q; x++) {
    B[v+x][u+w] = A[u+w][v+x];}}}}
```

Let us show indexes as (u,v,w,x)
Let Q (blocking factor) =2

Iteration 1. (0,0,0,0). We need A00 and B00. We access one block each from A and B

Cache	A00 A01	B00 **B01**		

2 miss

Iteration 2. (0,0,0,1). Need A01 and B10. A01 is cache-hit. B10 is cache miss

Cache	A00 A01	B00 B01	B10 B11	

1Hit 1Miss

$B[v+x][u+w] = A[u+w][v+x]$

Iteration 3. (0,0,1,0). Need A10 and B01. B01 is a cache hit. A10 is a cache miss

Cache	A00 A01	B00 B01	B10 B11	A10 A11

1 hit 1 miss

Iteration 4. (0,0,1,1). Need A11 and B11. Both are cache hits

Cache	A00 A01	B00 B01	B10 B11	A10 A11

2 Hits

Iteration 5. (0,2,0,0). Need A02 and B20. Both are miss. Fetch them.
Remove oldest blocks

2 miss

Observation 1: In 4 iterations, we get 4 hits and 4 misses. Out of 4 accesses to A, 2 are misses. Same for B

For whole matrix-transpose, we will get 16 misses and 16 hits

Interesting observation

Let us look at the cache contents at the end of iteration 4:

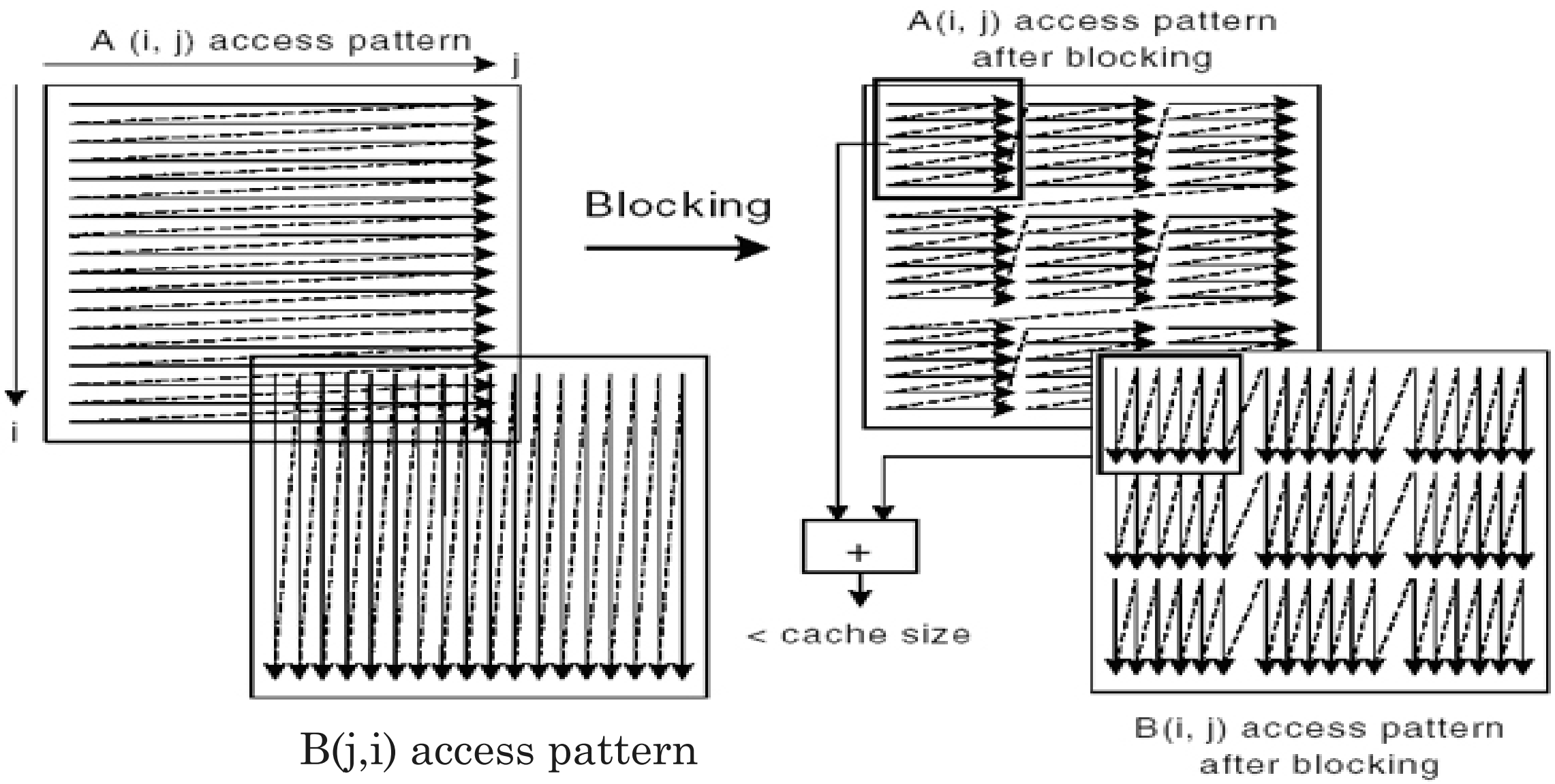
Cache				
	A00 A01	B00 B01	B10 B11	A10 A11

Clearly, cache has saved one tile from A and one tile from B.

B00	B01	B02	B03
B10	B11	B12	B13
B20	B21	B22	B23
B30	B21	B22	B23

A00	A01	A02	A03
A10	A11	A12	A13
A20	A21	A22	A23
A30	A21	A22	A23

This realization clearly explains the working of cache blocking (tiling).



B(j,i) access pattern

Summary

Design	Number of misses
No cache-blocking, B in column-major	16
No cache-blocking, B in row-major	24
Cache-blocking, B in row-major	16